

Project Case	
COMP6115 COMP6115001 Object-Oriented Analysis & Design	
Computer Science	O262-COMP6115-JK02-00
<i>Valid in Odd Semester Year 2025/2026</i>	Revision 00

1. Kelompok tidak diperkenankan untuk:

Members of the group are prohibited from:

- Melihat sebagian atau seluruh jawaban kelompok lain,
Seeing part or the whole answer from other groups,
- Menyadur sebagian atau seluruh jawaban dari buku, catatan, video, dan jenis referensi lainnya,
Retell a part or the whole answer from books, notes, videos, and other references,
- Menyadur sebagian atau seluruh jawaban dari internet,
Retell a part or the whole answer from the internet,
- Mengumpulkan jawaban yang tidak sesuai dengan tema soal,
Submitting an answer with a different theme from the given case,
- Melakukan tindakan yang menyebabkan jawaban dicontek oleh orang lain atau kelompok lain, baik disengaja maupun tidak disengaja,
Doing actions that could result in the answer being copied by someone or other groups, intentionally or unintentionally,
- Melakukan tindakan kecurangan lainnya.
Committing other dishonest actions.

2. Jika kelompok terbukti melakukan tindakan seperti yang dicantumkan pada butir ke-1, maka nilai mahasiswa dan/atau kelompok yang melakukan kecurangan, baik menyontek maupun dicontek, akan dinolkan sesuai dengan peraturan yang berlaku.

If it has been proven that a group has committed dishonest actions outlined in point 1 above, the whole group related to the incident, regardless of which one copies or has their answer copied, will be issued a score of zero according to the regulation.

3. Jawaban yang dapat diterima dan dinilai adalah jawaban yang dikumpulkan sebelum batas waktu yang telah ditentukan.

The answer must be submitted before the designated deadline to be accepted and graded,

4. Jawaban akan dinilai berdasarkan teknik atau metode yang diajarkan pada kelas praktikum dengan menggunakan software yang sudah ditentukan.

The scoring will be based on the materials taught during the practicum classes using the designated software. Using different software than requested may result in your answer not being graded.

5. Jika Anda tidak membaca peraturan ini, maka Anda dianggap sudah membaca dan menyetujuinya.

By taking this exam, you agree to these regulations, regardless of whether you have read them or not.

6. Persentase penilaian untuk mata kuliah ini adalah sebagai berikut:

The score will be distributed as follows:

Tugas Mandiri <i>Assignment</i>	Proyek <i>Project</i>	UAP <i>Final Exam</i>
40%	60%	-

7. Perangkat lunak yang digunakan pada mata kuliah ini adalah sebagai berikut:

This course uses the following software:

Software <i>Software</i>
Eclipse 2024.12 R Java 11.0.18 MySQL Java Connection Library 8.0.24 Visual Paradigm Community Edition 17.1 XAMPP 8.2.12

8. Ekstensi file yang harus dikumpulkan untuk mata kuliah ini adalah sebagai berikut:

Your answers must be in the following file extensions:

Tugas Mandiri <i>Assignment</i>	Proyek <i>Project</i>	UAP <i>Final Exam</i>
VPP, Image Files	JAVA, CLASS, SQL	-

9. File yang harus dikumpulkan adalah keseluruhan jawaban beserta dengan aset yang digunakan (gambar, audio, video, dll.) dan dokumentasi proyek yang berisikan link referensi aset dan penjelasan mengenai aplikasi yang dibuat (terlampir bersama dengan soal).

*Include other files that can support your project, such as: all files in your project, other files (image, audio, video, etc.) used in your project, *.doc file (documentation of your project) that contains all pages in your project, reference links of additional files (image, audio, video, etc.) used in your project, the description about how to use your application, etc.*

Soal

Case

JoymarKet

JoymarKet is a digital marketplace platform focusing on fresh food, meats, and groceries, delivered instantly to customers' doors. Due to the growing demand for digital convenience and the increasing competition in the e-grocery segment, JoymarKet needs a comprehensive and modern system to manage its operations. Below is the business flow and functional scope that must be covered by the system.

- **Customer Register**

The *customer* can create a new account by providing identity and contact information. Validation ensures correctness and prevents duplicate accounts.

Variable	Validation
Customer Id (update)	- Must exist. - Must be unique. - Cannot be changed.
Customer Full Name	- Cannot be empty.
Customer Email	- Must be filled. - Must end with @gmail.com. - Must be unique.
Customer Password	- Must be at least 6 characters.
Customer Confirm Password	- Must match Password.
Customer Phone	- Must be filled. - Must be numeric. - Must be 10–13 digits.
Customer Address	- Must be filled
Customer Gender	- Must be chosen

- **Customer Top Up Balance**

The *customer* can increase their account balance through a top-up form. The amount must be valid and meet the minimum threshold.

Variable	Validation
Amount	- Must be filled. - Must be numeric.

	- Minimum 10,000.
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• Customer Add Product to Cart

The *customer* can add products to their shopping cart. Validation ensures the count is within available stock.

Variable	Validation
Count	- Must be filled. - Must be numeric. - Must be between 1 and available stock.

• Customer Checkout

The *customer* can proceed to place an order after reviewing the cart. If a promo code is entered, it must exist. The customer's balance must be sufficient.

Variable	Validation
Promo	- If filled, must exist in DB.
Balance Check	- Must be greater than or equal to Order Total (from DB)

• Customer Edit Profile

Any *logged-in user* can update their personal profile details such as name, address, and phone number.

Variable	Validation
Full Name	- Must be filled.
Phone	- Must be numeric and 10–13 digits
Address	- Must be filled

• Admin Edit Product Stock

The *admin* is responsible for managing product stock quantities. The new stock value must be valid and non-negative.

Variable	Validation
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Stock	- Cannot be negative - Must be numeric
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• Admin Assign Order to Courier

The *admin* can assign a courier to an order that is ready for delivery. The selected courier must exist in the database.

Variable	Validation
Courier	- Must be selected. - Must exist in the database.

• Courier Edit Delivery Status

The *courier* can update the delivery status of an assigned order. The status must match one of the allowed values.

Variable	Validation
Status	- Must be selected (Pending / In Progress / Delivered).

• Update Cart Item

The *customer* can change the count of items in their cart. The new count must be within stock limits.

Variable	Validation
Count	- Must be filled. - Must be numeric. - Must be between 1 and available stock.

• Login

All users must log in before accessing the platform. Their credentials are matched against stored values in the database.

Email	- Must be filled. - Must match an email in the database.
Password	- Must be filled. - Must match the password in the database.

To help you design the program, JoymarKet has already hired Mrs. Caroline, a System Analyst, to provide you with the design of the program based on the requirements and the business flow above. You can open “**diagram.vpp**” to view the diagram that was designed by Mr. Budi. As a programmer, Hestia wants you to develop the application with the following requirements:

- The application must be built using Java-based Programming with **MVC (Model-View-Controller)** architecture for a better development process.
 - **Model**
The model layer is responsible for **representing concepts** in the business or information about the business situation. Besides that, the model layer is also responsible for **giving access to the database** via its public interfaces to acquire and manipulate references to preexisting domain objects.
 - **View**
View layer, or Presentation Layer, is responsible for showing information to the user and interpreting the user's commands. This layer is the home for **all user interfaces** in the project.
 - **Controller**
This layer is responsible for validating all input from the view layer, and **all business logic** is implemented in the controller layer. It is also responsible for **delegating requests** from the user to the lower layer for further processing.
- Ensure that **you didn't use regex, Java FXML, JavaFX Scene Builder, or any library used outside the class materials** in your project, or **it will affect your score**.
- The database must use **MySQL**.
- The application must have an **authenticated user based on roles**.
- The application must **minimize human error with a great user experience**.
- The application **must be made based on the analysis diagram** that was provided by Mrs. Caroline, but you may add additional components to the system based on your assumptions.
- Documentation for the application:
 - The guide for using the application in the form of **DOCX (external)**.
 - Explanation for your code in the form of **comments (internal)**.
 - Any additional assumptions that you make to develop the program.

Please ask your teaching assistant if there are any related questions.